

# Summer 1985

**Problem 1 (Su85).** 1. Show that a real  $2 \times 2$  matrix  $A$  satisfies  $A^2 = -I$  if and only if

$$A = \begin{pmatrix} \pm\sqrt{pq-1} & -p \\ q & \mp\sqrt{pq-1} \end{pmatrix}$$

where  $p$  and  $q$  are real numbers such that  $pq \geq 1$  and both upper or both lower signs should be chosen in the double signs.

2. Show that there is no real  $2 \times 2$  matrix  $A$  such that

$$A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 - \varepsilon \end{pmatrix}$$

with  $\varepsilon > 0$ .

**Problem 2 (Su85).** 1. For  $0 \leq \theta \leq \frac{\pi}{2}$ , show that  $\sin \theta \geq \frac{2}{\pi} \theta$ .

2. By using Part 1, or by any other method, show that if  $\lambda < 1$ , then

$$\lim_{R \rightarrow \infty} R^\lambda \int_0^{\pi/2} e^{-R \sin \theta} d\theta = 0$$

**Problem 3 (Su85).** Let  $A$  be a nonsingular real  $n \times n$  matrix. Prove that there exists a unique orthogonal matrix  $Q$  and a unique positive definite symmetric matrix  $B$  such that  $A = QB$ .

**Problem 4 (Su85, Fa21).** Let  $G$  be a group of order 120, let  $H$  be a subgroup of order 24, and assume that there is at least one left coset of  $H$  (other than  $H$  itself) which is equal to some right coset of  $H$ . Prove that  $H$  is a normal subgroup of  $G$ .

**Problem 5 (Su85, Fa19).** By the Fundamental Theorem of Algebra, the polynomial  $x^3 + 2x^2 + 7x + 1$  has three zeros,  $\alpha_1$ ,  $\alpha_2$ , and  $\alpha_3$ . Compute  $\alpha_1^3 + \alpha_2^3 + \alpha_3^3$ .

**Problem 6 (Fa79, Fa80, Sp85, Su85, Fa96, Fa98, Sp99, Sp26).** Prove that

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x} dx = \frac{\pi}{\sin \pi \alpha}$$

What restrictions must be placed on  $\alpha$ ?

**Problem 7 (Su85).** Let  $f(x) = e^{x^2/2} \int_x^\infty e^{-t^2/2} dt$  for  $x > 0$ .

1. Show that  $0 < f(x) < \frac{1}{x}$ .
2. Show that  $f(x)$  is strictly decreasing for  $x > 0$ .

**Problem 8 (Su85).** Let  $f$  be a real valued continuous function on a compact interval  $[a, b]$ . Given  $\varepsilon > 0$ , show that there is a polynomial  $p$  such that  $p(a) = f(a)$ ,  $p'(a) = 0$ , and  $|p(x) - f(x)| < \varepsilon$  for  $x \in [a, b]$ .

**Problem 9 (Su85).** Let  $u(x)$ ,  $0 \leq x \leq 1$ , be a real valued  $\mathcal{C}^2$  function which satisfies the differential equation

$$u''(x) = e^x u(x)$$

1. Show that if  $0 < x_0 < 1$ , then  $u$  cannot have a positive local maximum at  $x_0$ . Similarly, show that  $u$  cannot have a negative local minimum at  $x_0$ .
2. Now suppose that  $u(0) = u(1) = 0$ . Prove that  $u(x) \equiv 0$ ,  $0 \leq x \leq 1$ .

**Problem 10 (Sp79, Su85, Sp89, Fa19).** Prove that if  $1 < \lambda < \infty$ , the function

$$f_\lambda(z) = z + \lambda - e^z$$

has only one zero in the half-plane  $\Re z < 0$ , and that this zero is real.

**Problem 11 (Su85).** 1. Let  $G$  be a cyclic group, and let  $a, b \in G$  be elements which are not squares. Prove that  $ab$  is a square.

2. Give an example to show that this result is false if the group is not cyclic.

**Problem 12 (Su85, Fa89).** Let  $A$  be an  $n \times n$  real matrix and  $A^\top$  its transpose. Show that  $A^\top A$  and  $A^\top$  have the same image.

**Problem 13 (Su85, Fa19).** Let  $P(z)$  be a polynomial of degree  $< k$  with complex coefficients. Let  $\omega_1, \dots, \omega_k$  be the  $k$ -th roots of unity in  $\mathbb{C}$ . Prove that

$$\frac{1}{k} \sum_{i=1}^k P(\omega_i) = P(0)$$

**Problem 14 (Fa83, Su85).** Let  $M_{n \times n}(\mathbb{F})$  be the ring of  $n \times n$  matrices over a field  $\mathbb{F}$ . For  $n \geq 1$  does there exist a ring homomorphism from  $M_{(n+1) \times (n+1)}(\mathbb{F})$  onto  $M_{n \times n}(\mathbb{F})$ ?

**Problem 15 (Su85).** For each  $k > 0$ , let  $X_k$  be the set of analytic functions  $f(z)$  on the open unit disc  $\mathbb{D}$  such that

$$\sup_{z \in \mathbb{D}} \left\{ (1 - |z|)^k |f(z)| \right\}$$

is finite. Show that  $f \in X_k$  if and only if  $f' \in X_{k+1}$ .

**Problem 16 (Su85, Fa96).** A function  $f : [0, 1] \rightarrow \mathbb{R}$  is said to be *upper semicontinuous* if given  $x \in [0, 1]$  and  $\varepsilon > 0$ , there exists a  $\delta > 0$  such that if  $|y - x| < \delta$ , then  $f(y) < f(x) + \varepsilon$ . Prove that an upper semicontinuous function  $f$  on  $[0, 1]$  is bounded above and attains its maximum value at some point  $p \in [0, 1]$ .

**Problem 17 (Su85).** Let

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

where all the  $a_n$  are nonnegative reals, and the series has radius of convergence 1. Prove that  $f(z)$  cannot be analytically continued to a function analytic in a neighborhood of  $z = 1$ .

**Problem 18 (Su79, Fa79, Fa82, Su85).** Find all pairs of  $C^\infty$  functions  $x(t)$  and  $y(t)$  on  $\mathbb{R}$  satisfying

$$x'(t) = 2x(t) - y(t) \quad y'(t) = x(t)$$

**Problem 19 (Su85).** Let  $A_1 \geq A_2 \geq \cdots \geq A_k \geq 0$ . Evaluate

$$\lim_{n \rightarrow \infty} (A_1^n + A_2^n + \cdots + A_k^n)^{1/n}$$

Note: See also ??.

**Problem 20 (Su85).** Let  $\mathbb{F}$  be a field of characteristic  $p > 0$ ,  $p \neq 3$ . If  $\alpha$  is a zero of the polynomial  $f(x) = x^p - x + 3$  in an extension field of  $\mathbb{F}$ , show that  $f(x)$  has  $p$  distinct zeros in the field  $\mathbb{F}(\alpha)$ .